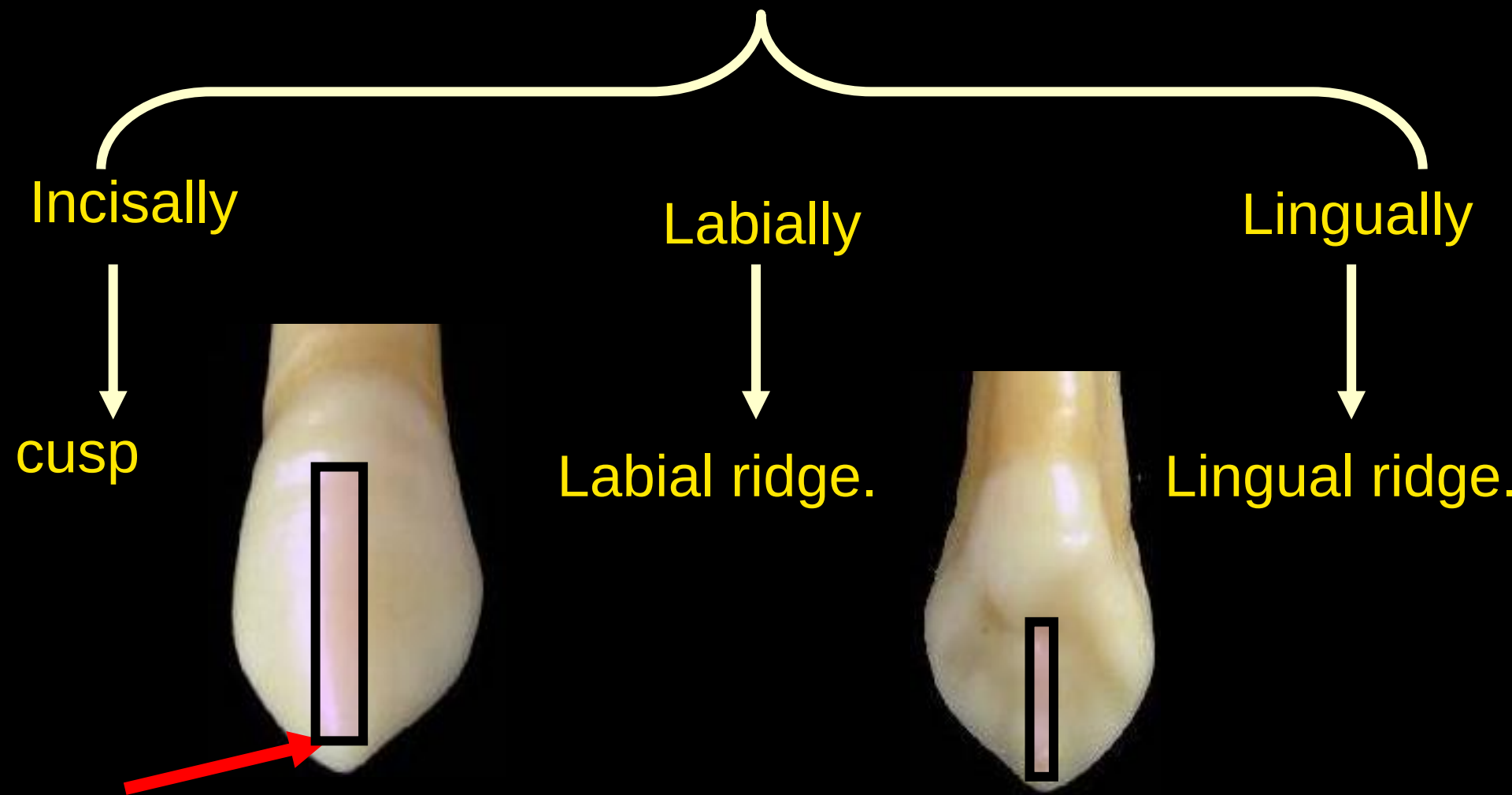


Canines



Number of lobes (four lobes 3 labially and 1 lingually).

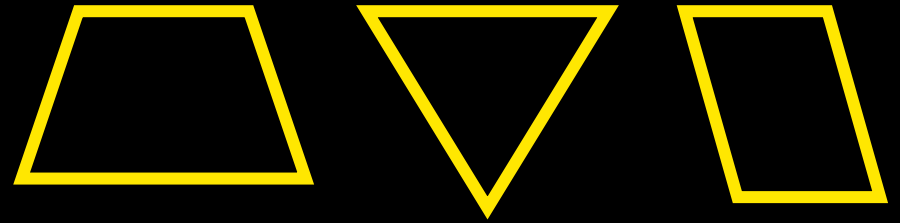
Note: the middle lobe of the canine is well developed



For proper tooth description

We have to speak about :

- *Geometric outline of the crown.*



- *Outline form of the crown and root.*

Convex



Concave



- *Surface anatomy of the crown and root (anatomical landmarks).*

General characteristics

- The canines are called **cuspid**s
- Longer & stronger roots
- Canine eminence:** the bone ridge over the labial portions of the roots. It has cosmetic value ensure normal facial expression

The upper canine has 5 surfaces:



Labial



Lingual



Mesial



Distal

Incisal



The lower canine has 5 surfaces



Labial



Lingual



Mesial



Distal

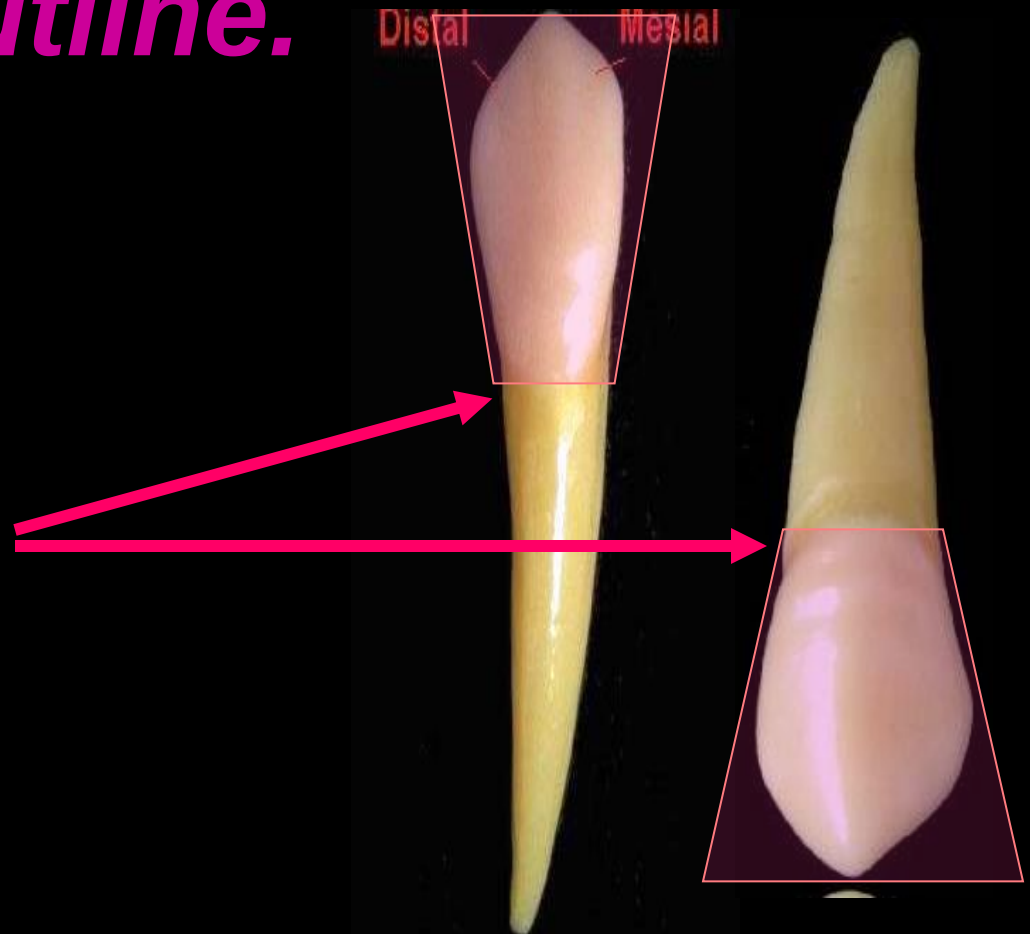


Incisal

Geometric outline of the crown

Labial and lingual surfaces have trapezoid outline.

The smallest uneven side cervically.



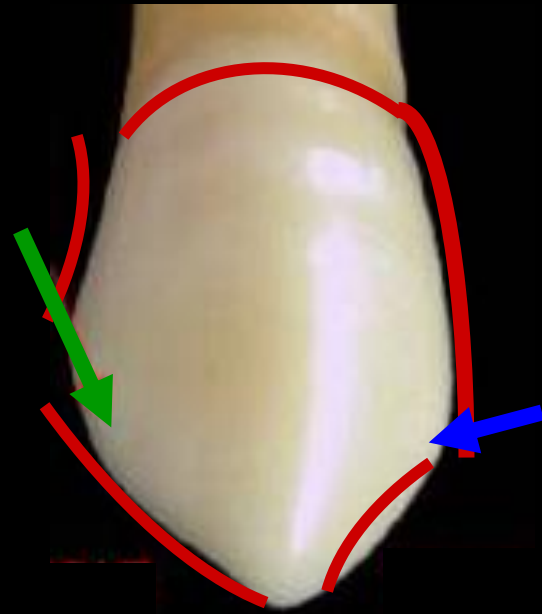
Labial surface of 3

Starting from the cervical line:

Mesial outline:, convex till the contact area (at the junction of I 1/3 and M 1/3) then continues as concave till the cusp tip.

Distal outline: concave till the contact area (at the middle third) then continues as convex till the cusp tip.

Cervical outline: convex root wise.

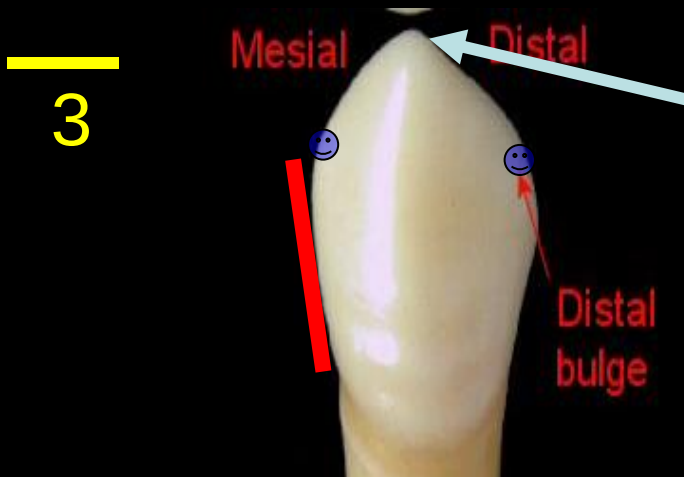


The mesial slope is shorter than the distal slope.

The cusp tip is pointed and on line with the long axis but slightly mesially deviated.



Labial surface of 3



Mesial outline is straight.

Both contact areas are more incisally so the crown appear longer and thinner than 3



Mesial outline is convex.

Contact areas are more cervically.

Outlines of the root



The mesial and distal outlines of the root is tapered to a distally curved apex

Surface anatomy of the crown and root.

Labial surface:

Elevations:

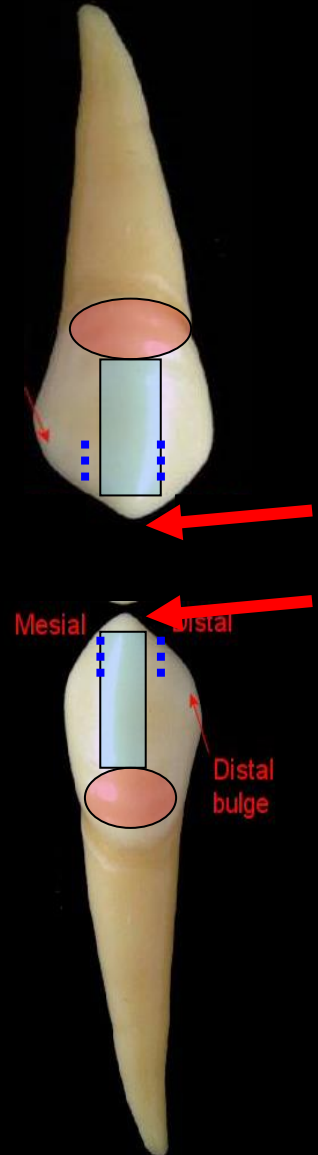
- It is convex with the maximum convexity at the cervical ridge.
- There is prominent ridge runs from the tip of the cusp toward the cervical margin (Labial ridge).

The middle lobe is well developed giving the cusp

Depressions:

Shallow longitudinal depressions lie mesial and distal to the labial ridge.

The root shows convex smooth surface



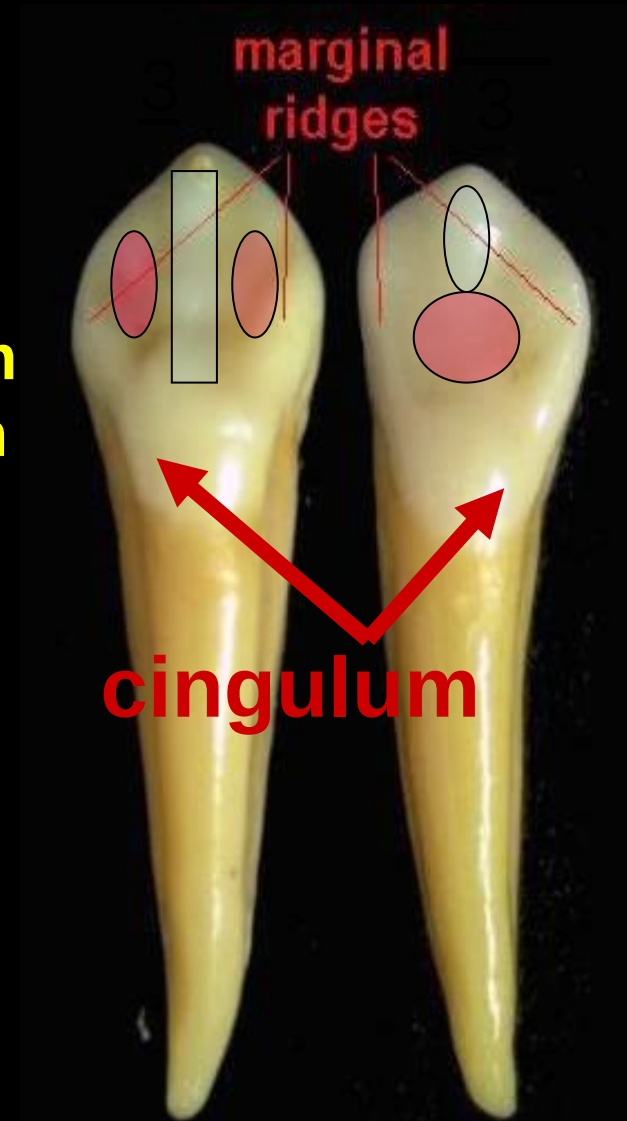
Lingual surface

Elevations:

- Marginal ridges, cingulum,
- Prominent lingual ridge that extend from the cusp tip till the cingulum in 3 while in $\overline{3}$ it's poorly developed & might be restricted to the incisal third.

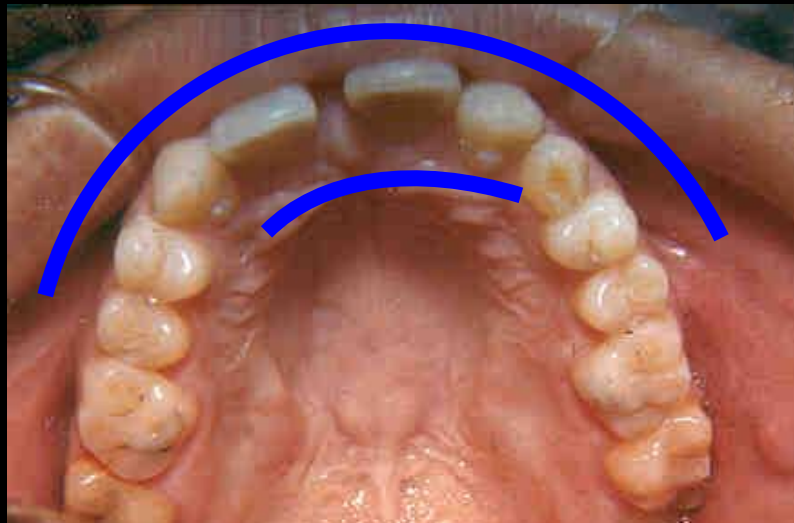
Depressions:

Lingual fossa that is divided into two fossae in 3 but might be one in $\overline{3}$.



Note:

- The elevations of the lower canine are not well developed as in case of the upper.
- The lingual surfaces in both are narrower in size than the labial surface due to the lingual convergence to accommodate

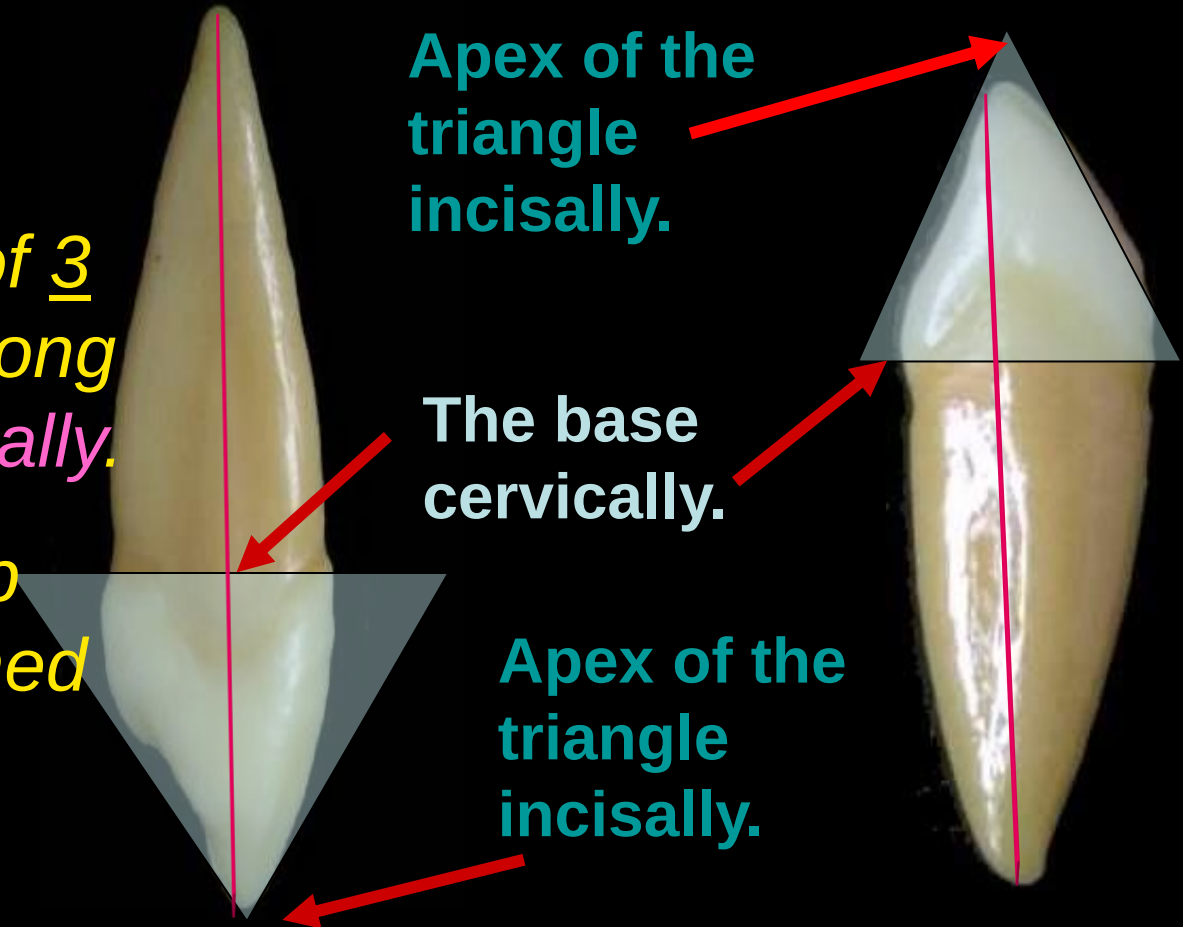


Proximal surfaces of the crown

The geometric outlines are triangular shape

Note: the cusp tip of 3 centralized on the long axis or inclined labially.

while the 3 cusp tip centralized or inclined lingually.



The outline form of the crown

Labial outline is convex with the crest of curvature at the cervical third representing

Lingual outline is convex cervically representing...

In the middle it is **straight** due to the presence of the ridge

In the incisal third it's convex again representing....





Cervical outline: curved incisally however, this curvature is less on the distal than the mesial.

Notes:

- Lingual outline in the middle of $\bar{3}$ is concave rather than straight due to short lingual ridge.

Outlines of the root

Upper canine:

The outlines tapered from the cervical line to a blunt apex. It's apical third may curve labially.

Lower canine:

The outlines are straight and nearly parallel from the cervical line to the middle third then they taper to a more pointed apex



Surface anatomy of the crown and root

The crown surface is convex and smooth, except the area **cervical** to the contact area shows flattening (mesial surface) or concavity (distal surface).

Position of contact areas vary from mesial to distal of the same tooth and vary from upper to lower canine.....identify their positions as mentioned before.



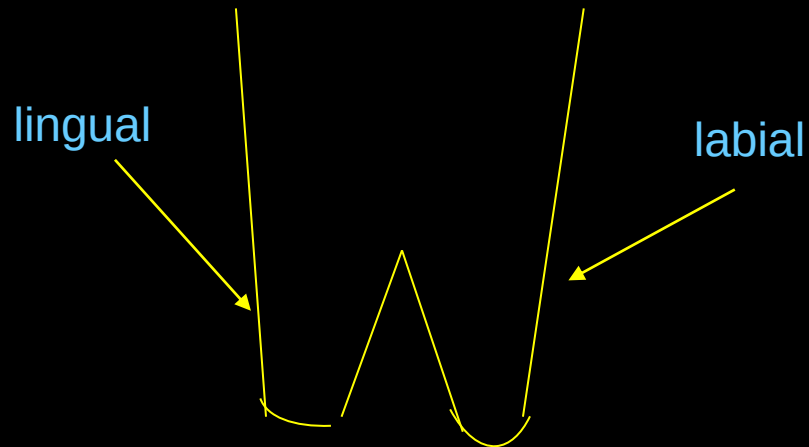
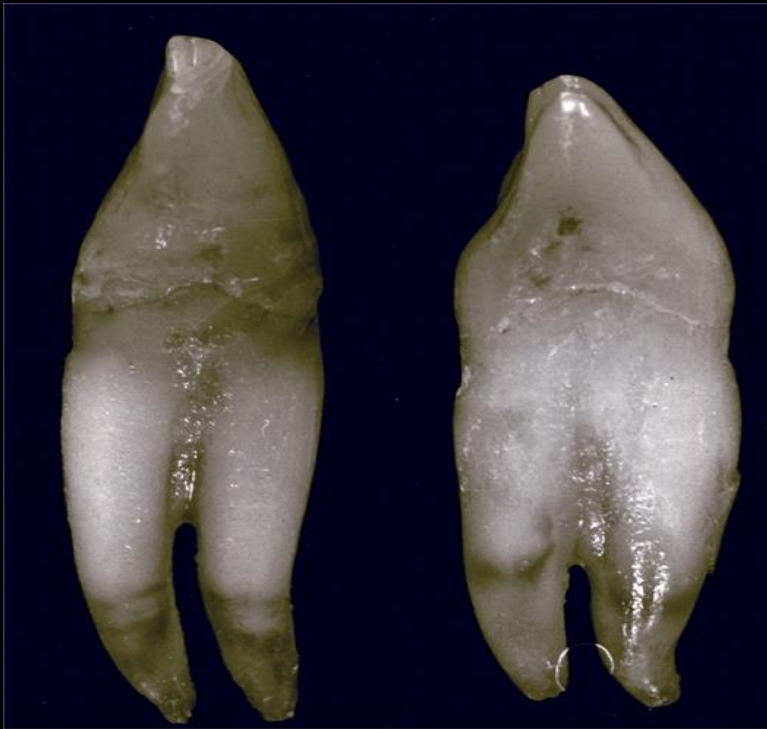
The root surface is broad with longitudinal depression.

The depression is shallower mesially than distally in the same tooth and is shallower in upper canine than in the lower canine.



Note: the depression in lower canine may be so deep causing bifurcation in the root. The bifurcation may be apically or extend up to cervical third.

If two roots exist they will be divided labially and lingually.



Incisal aspect.

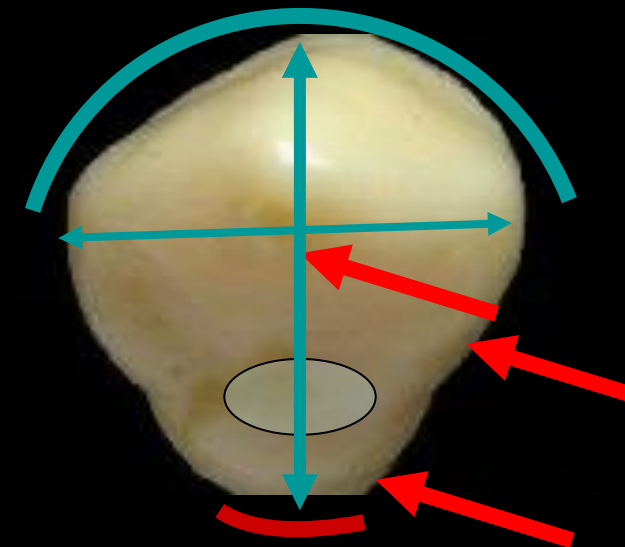
Outline and surface anatomy

The thickness is greater than the width.

The labial surface appear convex and even more than the incisors

The cingulum forms a short arc.

The elevations and depressions in the labial and lingual surfaces appear in this aspect



Pulp cavity.

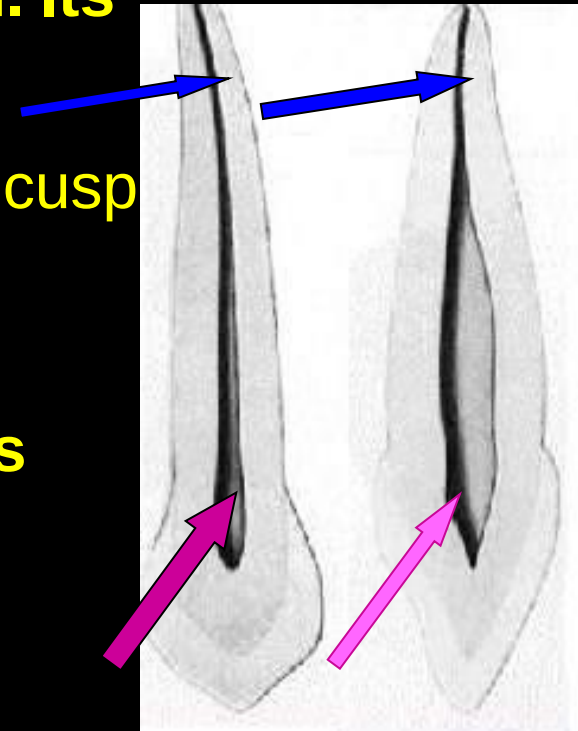
Pulp cavity is formed of:

Pulp chamber that is present in the crown. Its outline follows the outline of the crown.

In young teeth, it has pulp horn related to the cusp

Root canal presents in the root and follows its outline. The number of root canals in canines is only one.

Sometimes $\overline{3}$ has two root canals labially and lingually. The canals open in one apical foramen or separate foramina.



Thank You